DHARMSINH DESAI UNIVERSITY, NADIAD
FACULTY OF TECHNOLOGY
B.TECH SEMESTER-3 (CE)
FIRST SESSIONAL
SUBJECT: (CE-308) Design of Digital Circuits

Examination	: First Sessional	Seat No.	: CE113
Date	: 03/08/2022	Day	: Wednesday
Time	: 12.45 to 2 PM	Max. Marks	: 36

INSTRUCTIONS:

1. Figures to the right indicate maximum marks for that question.
2. The symbols used carry their usual meanings.
3. Assume suitable data, if required & mention them clearly.
4. Draw neat sketches wherever necessary.

Q.1 Do as directed. [12]

CO2 (a) Why should a digital circuit be minimized before implementing it? [2]

CO2 (b) How many total functions can be derived for 3 and 4 input variables? [1]

CO2 (c) Which one of the following is NOT a valid identity? [1]

I. $(x + y) \oplus z = x \oplus (y + z)$

II. $(x \oplus y) \oplus z = x \oplus (y \oplus z)$

III. $x \oplus y = x + y$, if $xy = 0$

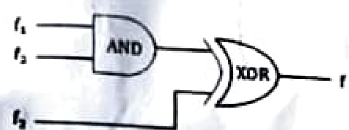
IV. $x \oplus y = (xy + x'y')'$

CO2 (d) Consider three 4-variable functions f_1, f_2 and f_3, which are expressed in sum-of-minterms as [2]

$f_1 = \Sigma(0, 2, 5, 8, 14)$, $f_2 = \Sigma(2, 3, 6, 8, 14, 15)$, $f_3 = \Sigma(2, 7, 11, 14)$

For the following circuit with one AND gate and one XOR gate, the output function f can be expressed as:

$f = \Sigma\ (?)$



CO1 (e) Convert decimal 225.225 to binary, octal and hexadecimal. [2]

CO1 (f) Write rules to perform subtraction operation M – N using r's complement. [2]

CO2 (g) Draw the logic diagram of realization of AND and OR Gate using NOR gate. [2]

Q.2 Attempt *Any TWO* from the following questions. [12]

CO2 (a) Implement the following function with either NAND or NOR Gates. Use only four gates. Only the normal inputs are available. [6]

$F(w,x,y,z) = w'xz + w'yz + x'yz' + wxy'z$

$d = wyz$

CO2 (b) With the use of maps, find the simplest form in sum of products of the function F = fg, where f and g are given by [6]

$f(w,x,y,z) = wxy' + y'z + w'yz' + x'yz'$

$g(w,x,y,z) = (w + x + y' + z')(x' + y' + z)(w' + y + z')$

CO1 (c)

1. Obtain the weighted binary code and the odd parity bit generated for the base-12 digits using weights of 5421. [4]
2. Perform subtraction using 9's complement: 753-864 [2]

Q.3 Attempt the following [12]

CO2 (a) Determine Prime-implicants of the function using tabulation method: $F(w, x, z, y) = \Sigma(0,1,6,7,8,9,10,11,12,15)$ [6]

CO2 (b) I. Consider the Boolean operator with the following properties: [2]

$x\#0 = x$, $x\#1 = \bar{x}$, $x\#x = 0$ and $x\#\bar{x} = 1$.

Determine equivalent expression for x#y.

II. Convert the following function in terms of maxterms using Boolean Algebra [4]

$F(A, B, C, D) = (D' + AB' + A'C)' + AC'D$

OR

Q.3 Attempt the following [12]

CO2 (a) Convert Following to the minimal SOP and minimal POS form.

I. $F(x,y,z) = \sum(1,3,7)$ [3]

II. $F(A, C, D, B) = \sum(0,2,6,11,13,14)$ [3]

CO2 (b) Given the Boolean function: $F = xy + x'y' + y'z$

I. Implement it with AND, OR and NOT gates. [2]

II. Implement it with only OR and NOT gates. [2]

III. Implement it with only AND and NOT gates. [2]

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FACULTY OF TECHNOLOGY
B.TECH SEMESTER-3 (CE)
SECOND SESSIONAL EXAM
SUBJECT: (CE-308) Design of Digital Circuits

Examination	: Second Sessional	Seat No.	: CE113
Date	: 07/09/2022	Day	: Wednesday
Time	: 10:45 am to 12:00 pm	Max. Marks	: 36

INSTRUCTIONS:
1. Figures to the right indicate maximum marks for that question.
2. The symbols used carry their usual meanings.
3. Assume suitable data, if required & mention them clearly.
4. Draw neat sketches wherever necessary.

Q.1 Do as directed. **[12]**

CO2 (a) How many full-adders and half-adders are required to add 16-bit numbers? Justify your answer. [2]

CO2 (b) What are bubbled OR and bubbled AND equivalent to? What are they used for? [2]

CO2 (c) What is a race around condition? Which are the ways to eliminate it? [2]

CO2 (d) What is the difference between a normal and priority encoder? [2]

CO2 (e) Differentiate: Ripple carry adder and Look-ahead carry adder. [2]

CO2 (f) Implement using Multiplexer
$F(x,y) = x' + y$ [2]

Q.2 Attempt **Any TWO** from the following questions. **[12]**

CO2 (a) Design an optimum combinational circuit with three inputs, x, y, and z, and three outputs, A, B, and C.
When the binary input is 0, 1, 2, or 3, the binary output is one greater than the input. When the binary input is 4, 5, 6, or 7, the binary output is one less than the input. [6]

CO2 (b) What is a parity bit? Design 4 bit even parity generator. Use minimum possible gates to implement the same. [6]

CO3 (c) What is a flip-flop? What is an indeterministic state in a flip-flop? Which inputs cause it in case of NOR gates? Do all the flip-flops have an indeterministic state? [6]

Q.3 Attempt the following **[12]**

CO2 (a) Design a 4 line to 2 line priority encoder. Include an output E to indicate that at least one input is a 1 [6]

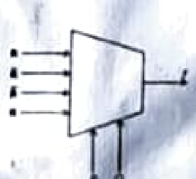
CO2 (b) 1. Construct a 5 X 32 decoder with four 3 X 8 decoders with enable and one 2 x 4 decoder. Use block diagrams. [3]
2. Derive the PLA programming table for the BCD to excess-3 code convertor. [3]

OR

Q.3 Attempt the following **[12]**

CO2 (a) 1. Implement below function using 4:1 Multiplexer.
$f(x, y, z) = m_0 + m_1 + m_3 + m_4 + m_5 + m_6$, where m_i is the i^{th} minterm. [3]

2. [3]



Consider above 4:1 Multiplexer circuit. Which of the below function is implemented by the above circuit. Justify your answer.
I. $P \oplus Q \oplus R$
II. $(P \oplus Q \oplus R)'$
III. $P+Q+R$
IV. $(P + Q + R)'$

CO2 (b) Design a combinational circuit that generated the 9's complement of a BCD digit. [6]

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FACULTY OF TECHNOLOGY
B.TECH SEMESTER-3 (CE)
THIRD SESSIONAL
SUBJECT: (CE-308) Design of Digital Circuits

Examination	: Third Sessional	**Seat No.**	: CE113
Date	: 12/10/2022	**Day**	: Wednesday
Time	: 9 AM to 10.15 AM	**Max. Marks**	: 36

INSTRUCTIONS:

1. Figures to the right indicate maximum marks for that question.
2. The symbols used carry their usual meanings.
3. Assume suitable data, if required & mention them clearly.
4. Draw neat sketches wherever necessary.

Q.1 **Do as directed.** [12]

CO3 R (a) The number of flip flops that will be complemented in a 10-bit binary ripple counter to reach the next count after the count 1 0 1 1 0 0 1 1 1 1 will be ______ [1]

CO2 U (b) The terminal (last) count of a modulus-11 binary counter is ______. [1]

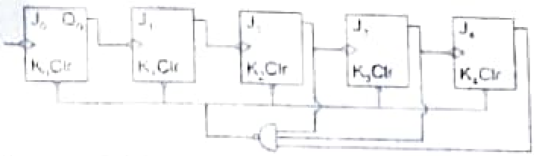
A (c) In order to use a shift register as a counter, ______. [1]

CO3
A. the register's serial input is the counter input and the serial output is the counter output
B. the parallel inputs provide the input signal and the output signal is taken from the serial data output
C. a serial-in, serial-out register must be used
D. the serial output of the register is connected back to the serial input of the register

(d) To generate 2^k timing signals, we can use either ______ number of flip flops or ______ bit counter with ______ dimension of decoder. [1]

A (e) What is the mod number of asynchronous counter shown (All J = K = 1) below. [2]

CO2

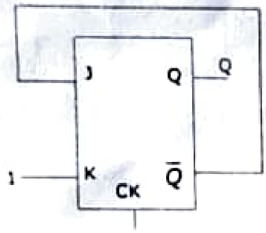


C (f) How many JK flip-flops will be required to implement the following counter?[Justify your answer] [2]

CO3
11 – 00 – 10 – 00 – 01 – 00 – 11 – 00 – 10 – 00 – 01 – 00 – 11 – 00 – 10 – 00 – 01 – 00 – ...

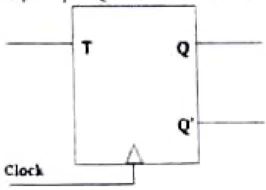
A (g) What will be the output for next 6 clock pulses in the following flip-flop? Assume that the flip-flop is currently in set state.[justify your ans] [2]

CO3



N (h) Q' of T flip-flop is zero. Clock transition from ____ to ____ when T = ____ will ensure flip-flop output Q to have zero value. [2]

CO3



(a) 0,1,1 (b) 0,1,0 (c) 1,1,1 (d) 1,0,1 (e) 1,1,0 (f) 0,0,0 (g) 0,0,1 (h) 1,0,0

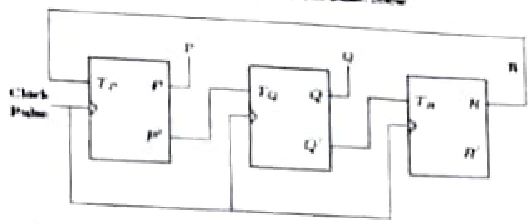
Q.2 Attempt ***Any TWO*** from the following questions. [12]

CO3 A (a) Design a Johnson counter for 10 timing signals along with the decoding gates. [6]

CO3 A (b) Design a Synchronous BCD Counter with JK flip-flops. [6]

	U	(c)	Consider a 3-bit counter, designed using T flip-flops, as shown below	[4]

CO3

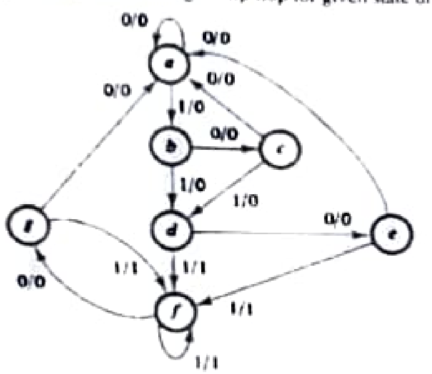


1. Assuming the initial state of the counter given by PQR as 000, what are the next three states?

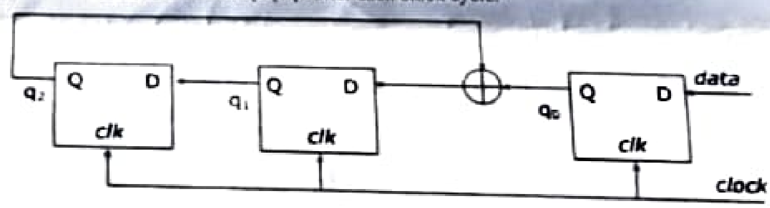
2. Draw the circuit of serial transfer using two 4 bit shift registers. [2]

Q.3 Attempt the following [12]

CO3 N (a) Design Logic diagram using D-flip flop for given state diagram. [6]



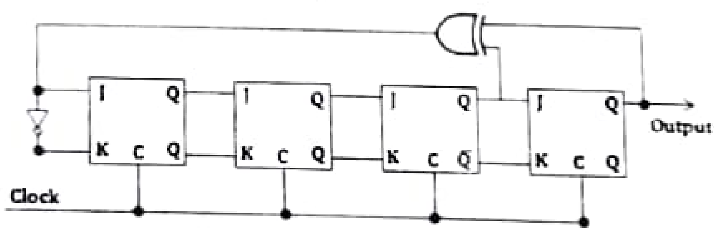
CO3 N (b) Consider the circuit in the diagram. The ⊕ operator represents Ex-NOR. The D flip flops are initialized to zeros (cleared). The data 10110001 is supplied to the "data" terminal in nine clock cycles. Find out the values of q2q1q0 after each clock cycle. [6]



OR

Q.3 Attempt the following [12]

CO3 N (a) Find out the output sequence of below given circuit. [6]



CO3 N (b) Design logic diagram using SR flipflop for below given circuit. [6]